

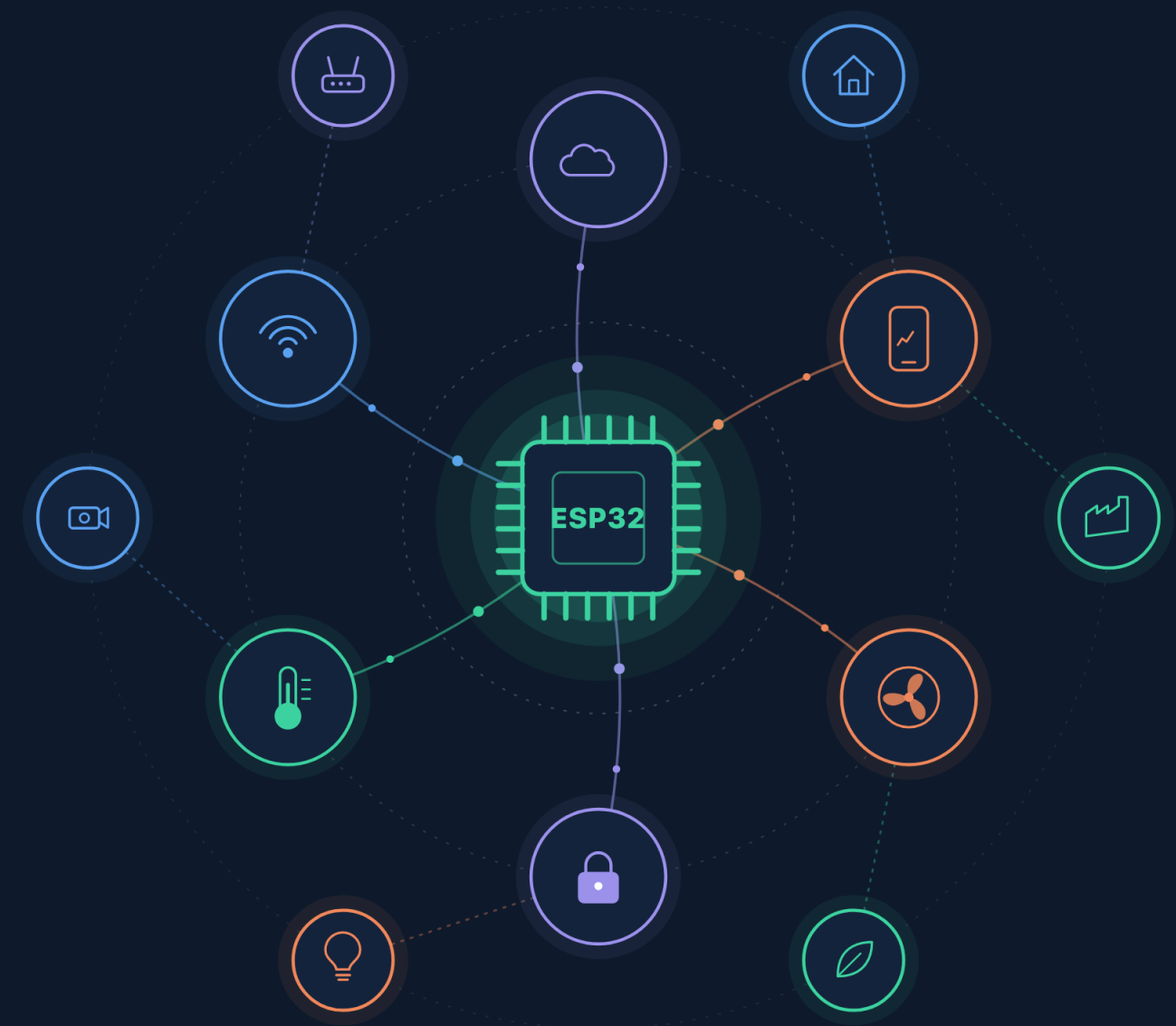
MODULE 6 • 45 MINUTES

Data Processing and Automation

How a device turns raw readings into decisions, and then acts on them by itself.

Mehdi Lotfinejad

Introduction to the Internet of Things



THE ROAD AHEAD

Eight modules, one smart device

PART 1 — DEVICES AND DATA

01

What is IoT?

Lecture

02

How it's built

Lecture

03

Sensors and actuators

Labs 1–2

04

Getting online

Lab 3

PART 2 — CLOUD, AUTOMATION AND SECURITY

05

Cloud and dashboards

Lab 4

06

Automation

You are here

07

Security

Group activity

08

Real world + capstone

Lab 6

BY THE END OF THIS MODULE

You will be able to...

-
- 01 **Describe** the steps data takes from raw reading to decision
 - 02 **Explain** what “event-driven” means, and why IoT uses it
 - 03 **Write** simple if-then rules, and explain what a rules engine does
 - 04 **Compare** automation at the edge and in the cloud
 - 05 **Build** a device that reacts to a temperature limit by itself
-

WHERE WE ARE

Your device can sense and send. Now it will decide and act, by itself.

A dashboard still needs a person. What if nobody is watching when the server room overheats at 3 am?

THE BIG PICTURE

From reading to action



Raw data

31.4, 31.6, 31.5 °C



Process

Clean, average and compare



Rule

Is it above 28 °C?



Act or wait

Yes: fan on + alert.
No: keep watching.

Three ideas: **processing, events and rules**, and **where the decision happens**.

Part A

From raw data to useful information

Raw sensor data is rarely ready to use as it is.

Seven steps from reading to decision

Step	What happens	Room monitor example
1. Collect	Read the sensor at a sensible rate	Read the DHT22 every 2 seconds
2. Validate	Throw away readings that can't be real	Ignore -40 °C or "not a number"
3. Clean / smooth	Reduce noise and random wobble	Average the last 5 readings
4. Transform	Convert units or combine values	"Feels like" from temperature + humidity
5. Aggregate	Summarise over time	Hourly average, daily maximum
6. Decide	Compare against rules	Is the average above 28 °C?
7. Store / act / send	Save it, act on it, or pass it on	Switch the fan, log it, send an alert

Not every system uses every step. But almost all do some version of **collect, check, clean, decide**.

One bad reading can cause a false alarm



Validate

A loose wire sends **85 °C**. Rule: ignore any reading more than 10 °C away from the last one.



Smooth

Near the limit, readings jump: 27.9, 28.1, 27.8, 28.2. Average a few before you decide.

PAUSE AND CHECK

Your sensor reports 24.1, 24.3, 24.2, 97.0, 24.2 °C. Which processing step should deal with the 97.0?

Look back at the seven steps. Think on your own for 30 seconds.

Part B

Events and rules

Noticing when something important happens, and deciding what to do about it.

Two ways to handle data



Polling

Check again and again:
“Temperature now? And
now?”



Event-driven

React when something
meaningful happens.



An event

A change that matters:
crossed 28 °C, door opened.

A child asking “Are we there yet?” every five minutes is polling. A driver saying “We’ve arrived!” is an event.

Why IoT systems are event-driven



Saves resources

One message when something changes, not 1,000 “nothing changed”.



Fast

React the moment it happens, not at the next check.



Scales

One event, many reactions: fan, alert, log, dashboard.

This is how **MQTT** works (Module 4): publish once, and every subscriber reacts.

AUTOMATION RULES

IF this, THEN that

IF (condition)	THEN (action)
Temperature > 28 °C	Switch the fan on
Motion detected AND after 10 pm	Switch on the porch light
Soil moisture < 20%	Open the irrigation valve for 5 minutes
No data for 15 minutes	Email the technician

Combine conditions with **AND** and **OR**. The service **IFTTT** is named after this pattern: “If This Then That”.

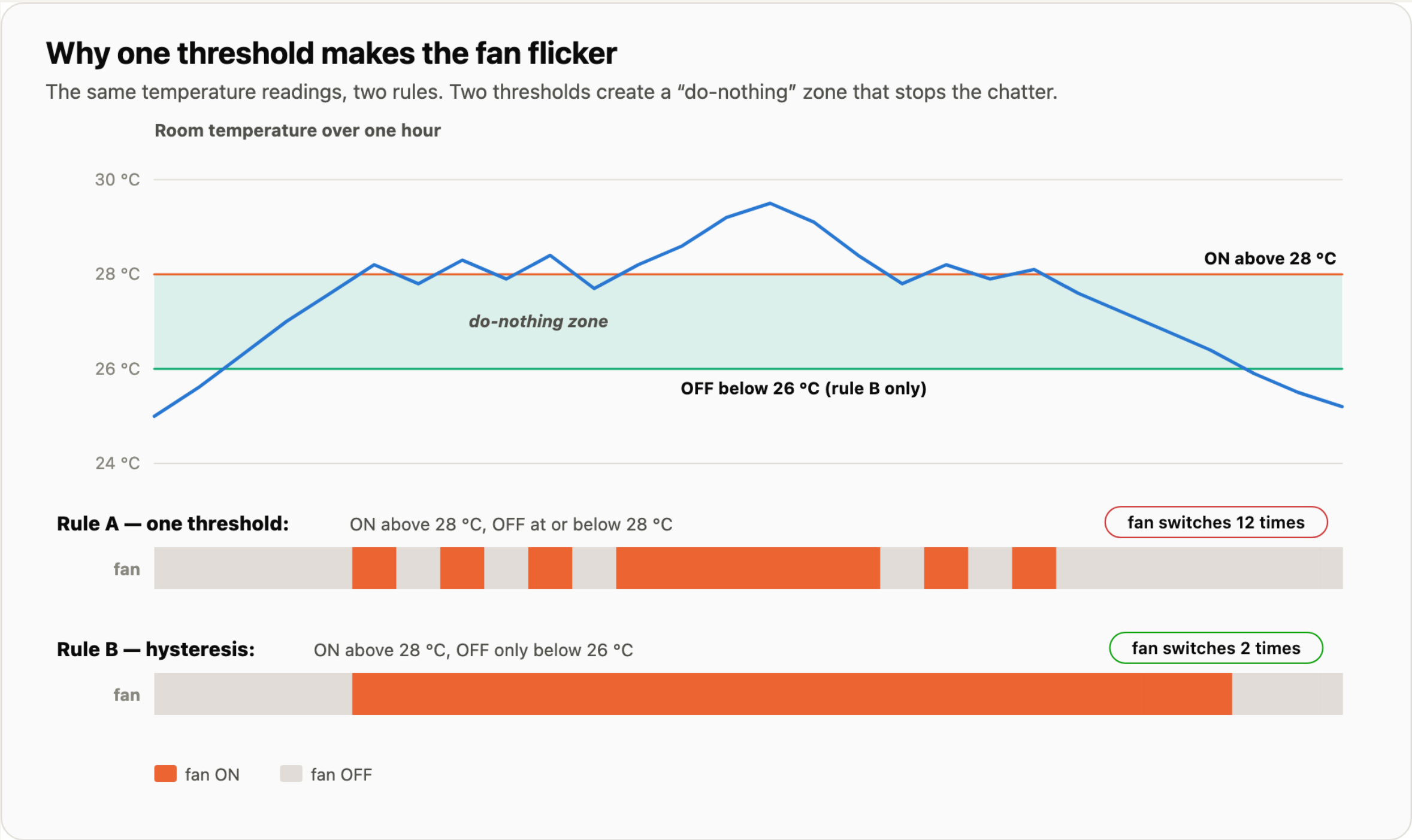
Rules engines in IoT platforms

Platform	Its rule feature, in plain terms
ThingSpeak	“React”: run an action when a channel’s value meets a condition
Blynk	“Automations”: if-then rules built in the app
AWS IoT Core	“Rules”: route and act on device messages as they arrive
Node-RED	Free, visual: build rules by connecting boxes with wires
Home Assistant	Free smart-home hub with automations for home devices

The idea is always the same as the “if” in Lab 5. Only where it runs and how you edit it change.

A CLASSIC RULE PROBLEM

Flickering, and the fix: hysteresis



PAUSE AND CHECK

Write a rule in plain English: alert if a freezer rises above -10°C , but only if it stays that warm for more than 10 minutes.

Use IF ... AND ... THEN. Why the 10 minutes? Talk to your neighbour for one minute.

Part C

Where should the decision happen?

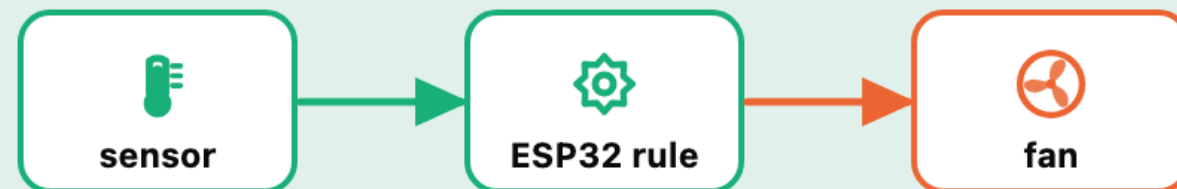
On the device, at the edge — or far away, in the cloud.

The same rule, two places to run it

The same rule, two places to run it

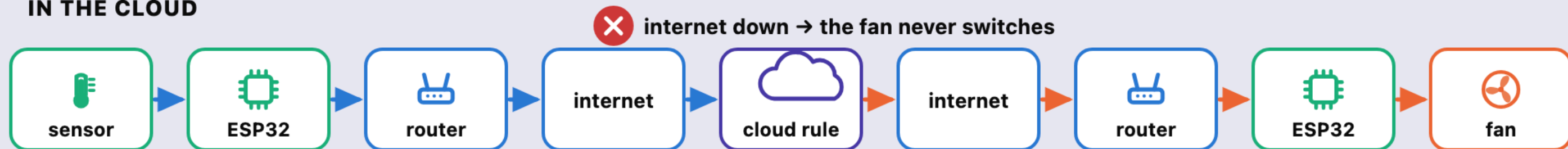
At the edge the decision takes two short hops. In the cloud it's a round trip — and it breaks if the internet does.

AT THE EDGE



✓ **≈ milliseconds · 2 hops**
keeps working with the internet down

IN THE CLOUD



✗ internet down → the fan never switches

≈ hundreds of milliseconds to seconds · 8 hops

but: easy to change for every device at once, and it sees data from all devices

Blue arrows: reading going up · orange arrows: command coming back

PROS AND CONS

Edge automation and cloud automation

	At the edge	In the cloud
Speed	Fast: a fraction of a second	Slower: a round trip over the internet
Internet down	Keeps working	Stops: no internet, no decision
Changing a rule	Update every device’s software	Edit once, applies to every device
What it sees	Only itself	Data from all devices
Computing power	Limited: a microcontroller	Lots, for heavy analysis and history
Data and cost	Sends less, saves money	Ongoing costs for all that data
Privacy	Sensitive data can stay on the device	More data sent, more privacy risk

CHOOSING, AND USING BOTH

Where should this rule run?

If the decision...	Put it...
Is urgent or safety-critical	At the edge
Must keep working without internet	At the edge
Needs data from many devices	In the cloud
Changes often, or is set by non-technical staff	In the cloud
Needs heavy analysis or long history	In the cloud

Most good systems use **both**: urgent, simple rules at the edge; big-picture rules in the cloud.

REAL-WORLD EXAMPLE

A cold-chain vaccine fridge



Processing

Validate, then average over five minutes.



Events

Left the safe range, power failed, offline.



Edge rules

A local buzzer at once, even with no internet.



Cloud rules

Text the pharmacist, escalate, keep a record.

Some vaccines must stay between **2 °C and 8 °C** at all times, even while they travel.

COMMON MISCONCEPTIONS

Four ideas that trip people up

✗ Myth *Automation means artificial intelligence.*

✓ Fact Most automation is simple if-then rules. AI builds on top.

✗ Myth *Edge and cloud compete. You must pick one.*

✓ Fact They are partners. Most systems split the work.

✗ Myth *Sending every reading to the cloud is safest.*

✓ Fact It costs more, drains batteries and fails offline.

✗ Myth *One threshold is enough for a good rule.*

✓ Fact It flickers. Use hysteresis or a time delay.

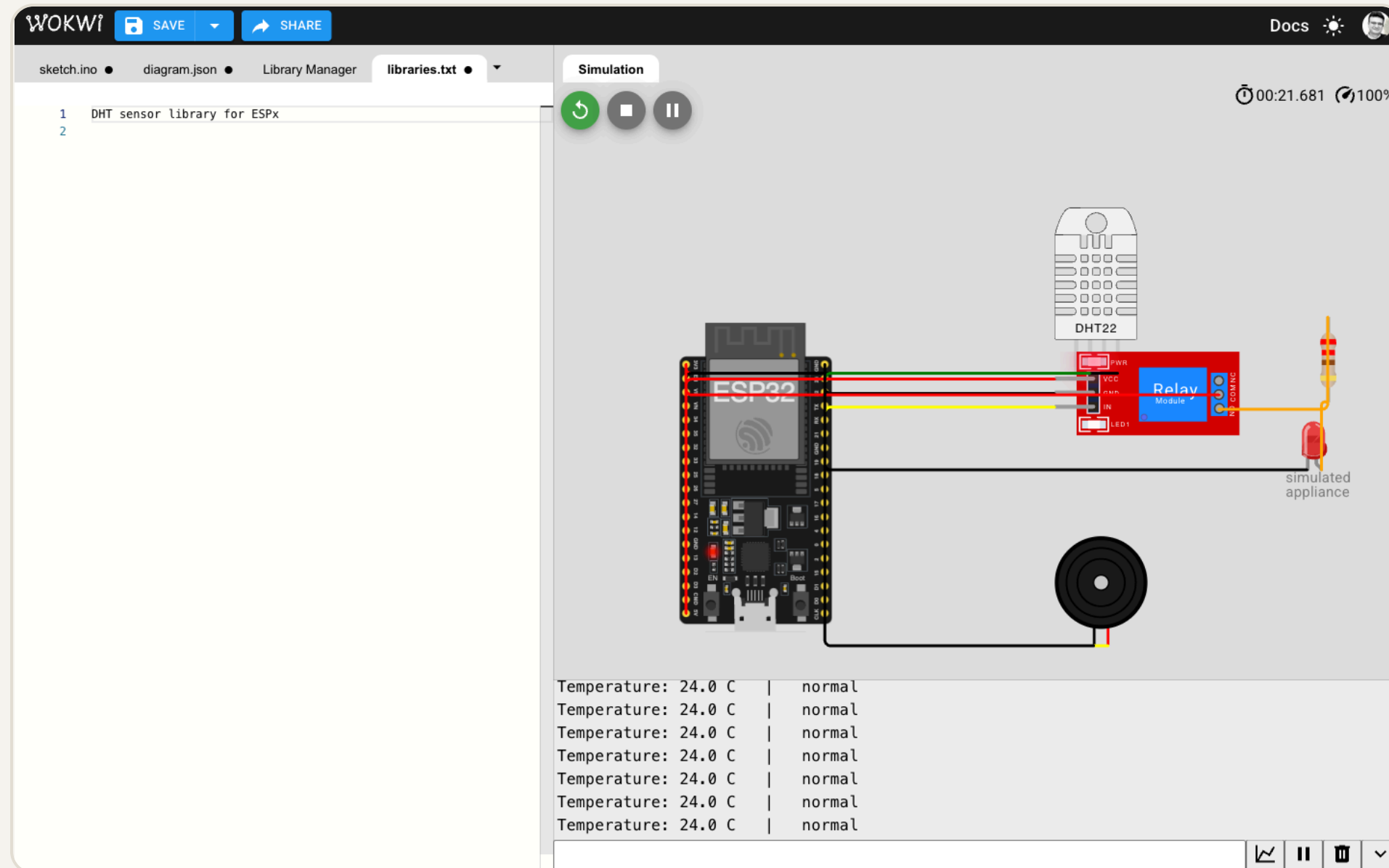
NEW PART FOR LAB 5

The relay module

- An **electrically controlled switch**
- Switches a much bigger circuit on or off
- The ESP32 drives it like an LED: one pin
- In Lab 5, an **LED stands in** for the fan



Automation: threshold-triggered relay



- 1 Open a new ESP32 project in Wokwi
- 2 Paste sketch.ino, diagram.json and libraries.txt
- 3 Press Play: a reading every 2 s, all “normal”
- 4 Click the DHT22 and drag the temperature above 28 °C
- 5 Watch the relay, LED and buzzer switch on by themselves



Scan for the step-by-step guide
github.com/datatweets/intro-to-iot-workshop

KEY TAKEAWAYS

What to remember

- ✓ Data flows through a **workflow**: collect, validate, clean, decide, act
- ✓ An **event** is a meaningful change; event-driven saves power and data
- ✓ Rules follow **IF** condition **THEN** action, with AND / OR
- ✓ A **rules engine** runs many rules you can change without code
- ✓ **Hysteresis** (on at 28 °C, off at 26 °C) stops flickering
- ✓ **Edge** is fast and works offline; **cloud** sees the big picture

Quick quiz

Q1 What's the difference between a reading and an event? Give an example of each.

Q2 A greenhouse fan switches on and off every few seconds. What's the problem, and the fix?

Q3 A gas-leak detector must shut a valve instantly, even with no internet. Edge or cloud?

Q4 Why would a large company use a rules engine instead of rules in device code?

UP NEXT • MODULE 7

IoT Security and Best Practices

We step back from building and look at what can go wrong: how IoT devices get attacked, and how real products defend against them.



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Book, labs and slides: github.com/datatweets/intro-to-iot-workshop